

Null It Out: Guarding Protected Attributes by Iterative Nullspace Projection

Shauli Ravfogel^{1,2} Yanai Elazar^{1,2} Hila Gonen¹ Michael Twiton³ Yoav Goldberg^{1,2}

¹Computer Science Department, Bar Ilan University

²Allen Institute for Artificial Intelligence

³Independent researcher

{shauli.ravfogel, yanaiela, hilagnn, mtwito101, yoav.goldberg}@gmail.com

Abstract

The ability to control for the kinds of information encoded in neural representation has a variety of use cases, especially in light of the challenge of interpreting these models. We present Iterative Null-space Projection (INLP), a novel method for removing information from neural representations. Our method is based on repeated training of linear classifiers that predict a certain property we aim to remove, followed by projection of the representations on their null-space. By doing so, the classifiers become oblivious to that target property, making it hard to linearly separate the data according to it. While applicable for multiple uses, we evaluate our method on bias and fairness use-cases, and show that our method is able to mitigate bias in word embeddings, as well as to increase fairness in a setting of multi-class classification.

1 Introduction

What is encoded in vector representations of textual data, and can we control it? Word embeddings, pre-trained language models, and more generally deep learning methods emerge as very effective techniques for text classification. Accordingly, they are increasingly being used for predictions in real-world situations. A large part of the success is due to the models' ability to perform *representation learning*, coming up with effective feature representations for the prediction task at hand. However, these learned representations, while effective, are also notoriously opaque: we do not know what is encoded in them. Indeed, there is an emerging line of work on probing deep-learning derived representations for syntactic (Linzen et al., 2016; Hewitt and Manning, 2019; Goldberg, 2019), semantic (Tenney et al., 2019) and factual knowledge (Petroni et al., 2019). There is also evidence that they capture a lot of information regarding the

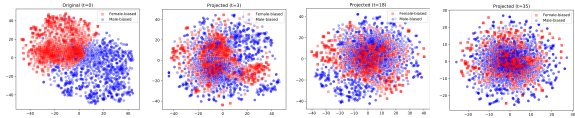


Figure 1: t-SNE projection of GloVe vectors of the most gender-biased words after $t=0, 3, 18$, and 35 iterations of INLP. Words are colored according to being male-biased or female-biased.

demographics of the author of the text (Blodgett et al., 2016; Elazar and Goldberg, 2018).

What can we do in situations where we *do not want* our representations to encode certain kinds of information? For example, we may want a word representation that does not take *tense* into account, or that does not encode *part-of-speech* distinctions. We may want a classifier that judges the *formality* of the text, but which is also oblivious to the *topic* the text was taken from. Finally, and also our empirical focus in this work, this situation often arises when considering *fairness* and *bias* of language-based classification. We may not want our word-embeddings to encode *gender* stereotypes, and we do not want sensitive decisions on hiring or loan approvals to condition on the *race*, *gender* or *age* of the applicant.

We present a novel method for selectively removing specific kinds of information from a representation. Previous methods are either based on projection on a pre-specified, user-provided direction (Bolukbasi et al., 2016), or on adding an adversarial objective to an end-to-end training process (Xie et al., 2017). Both of these have benefits and limitations, as we discuss in the related work section (§2). Our proposed method, Iterative Nullspace Projection (INLP), presented in section 4, can be seen as a combination of these approaches, capitalizing on the benefits of both. Like the projection methods, it is also based on the mathematical notion of linear projection, a commonly used de-

terministic operator. Like the adversarial methods, it is data-driven in the directions it removes: we do not presuppose specific directions in the latent space that correspond to the protected attribute, but rather *learn* those directions, and remove them. Empirically, we find it to work well. We evaluate the method on the challenging task of removing gender signals from word embeddings (Bolukbasi et al., 2016; Zhao et al., 2018). Recently, Gonen and Goldberg (2019) showed several limitations of current methods for this task. We show that our method is effective in reducing many, but not all, of these (§4).

We also consider the context of fair classification, where we want to ensure that a classifier’s decision is oblivious to a protected attribute such as race, gender or age. There, we need to integrate the projection-based method within a pre-trained classifier. We propose a method to do so in section §5, and demonstrate its effectiveness in a controlled setup (§6.2) as well as in a real-world one (§6.3).

Finally, while we propose a general purpose information-removal method, our main evaluation is in the realm of bias and fairness applications. We stress that this calls for some stricter scrutiny, as the effects of blindly trusting strong claims can have severe real-world consequences on individuals. We discuss the limitations of our model in the context of such applications in section §7.

2 Related Work

The objective of controlled removal of specific types of information from neural representation is tightly related to the task of disentanglement of the representations (Bengio et al., 2013; Mathieu et al., 2016), that is, controlling and separating the different kinds of information encoded in them. In the context of transfer learning, previous methods have pursued representations which are *invariant* to some properties of the input, such as genre or topic, in order to ease domain transfer (Ganin and Lempitsky, 2015). Those methods mostly rely on adding an adversarial component (Goodfellow et al., 2014; Ganin and Lempitsky, 2015; Xie et al., 2017; Zhang et al., 2018) to the main task objective: the representation is regularized by an adversary network, that competes against the encoder, trying to extract the protected information from its representation.

While adversarial methods showed impressive performance in various machine learning tasks, and

were applied for the goal of removal of sensitive information (Elazar and Goldberg, 2018; Coavoux et al., 2018; Resheff et al., 2019; Barrett et al., 2019), they are notoriously hard to train. Elazar and Goldberg (2018) have evaluated adversarial methods for the removal of demographic information from representations. They showed that the complete removal of the protected information is nontrivial: even when the attribute seems protected, different classifiers of the same architecture can often still succeed in extracting it. Another drawback of these methods is their reliance on a main-task loss in addition to the adversarial loss, making them less suitable for tasks such as debiasing pre-trained word embeddings.

Xu et al. (2017) utilized a “nullspace cleaning” operator for increasing privacy in classifiers. They remove from the input a subspace that contains (but is not limited to) the nullspace of a pre-trained classifier, in order to clean information that is *not* used for the main task (and might be protected), while minimally impairing classification accuracy. While similar in spirit to our method, several key differences exist. As the complementary setting – removing the nullspace of the main-task classifier vs. projection onto the nullspace of protected attribute classifiers – aims to achieve a distinct goal (privacy preserving), there is no notion of exhaustive cleaning. Furthermore, they do not remove protected attributes that *are* used by the classifier (e.g. when it conditions on gender).

A recent line of work focused on projecting the representation to a subspace which does not encode the protected attributes. Under this method, one identifies directions in the latent space that correspond to the protected attribute, and removes them. In a seminal work, Bolukbasi et al. (2016) aimed to identify a “gender subspace” in word-embedding space by calculating the main directions in a subspace spanned by the differences between gendered word pairs, such as $\vec{he} - \vec{she}$. They suggested to zero out the components of neutral words in the direction of the “gender subspace” first principle components, and actively pushed neutral words to be equally distant from male and female-gendered words. However, Gonen and Goldberg (2019) have shown that these methods only cover up the bias and that in fact, the information is deeply ingrained in the representations. A key drawback of this approach is that it relies on an intuitive selection of *a few* (or a single) gender directions, while, as we

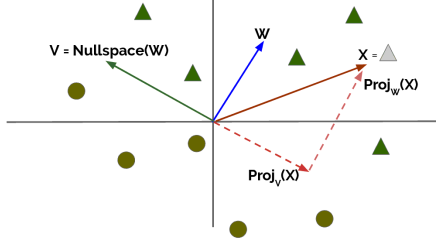


Figure 2: Nullspace projection for a 2-dimensional binary classifier. The decision boundary of W is W 's null-space.

reveal in our experiments, the gender subspace is actually spanned by dozens to hundreds of orthogonal directions in the latent space, which are not necessarily as interpretable as the *he* – *she* direction. This observation aligns with the analysis of [Ethayarajh et al. \(2019\)](#) who demonstrated that debiasing by projection is theoretically effective provided that one removes *all* relevant directions in the latent space.

3 Objective and Definitions

Our main goal is to “guard” sensitive information, so that it will not be encoded in a representation. Given a set of vectors $x_i \in \mathbb{R}^d$, and corresponding discrete attributes Z , $z_i \in \{1, \dots, k\}$ (e.g. race or gender), we aim to learn a transformation $g : \mathbb{R}^d \rightarrow \mathbb{R}^d$, such that z_i cannot be predicted from $g(x_i)$. In this work we are concerned with “linear guarding”: we seek a guard g such that no linear classifier $w(\cdot)$ can predict z_i from $g(x_i)$ with an accuracy greater than that of a decision rule that considers only the proportion of labels in Z . We also wish for $g(x_i)$ to stay informative: when the vectors x are used for some end task, we want $g(x)$ to have as minimal influence as possible on the end task performance, provided that z remains guarded. We use the following definitions:

Guarded w.r.t. a hypothesis class Let $X = x_1, \dots, x_m \in \mathcal{X} \subseteq \mathbb{R}^d$ be a set of vectors, with corresponding discrete attributes Z , $z_i \in \{1, \dots, k\}$. We say the set X is *guarded for Z with respect to hypothesis class $\mathcal{H}$* (conversely Z is *guarded in X*) if there is no classifier $W \in \mathcal{H}$ that can predict z_i from x_i at better than guessing the majority class.

Guarding function A function $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is said to be *guarding X for Z (w.r.t. to class $\mathcal{H}$)* if the set $\{g(x) | x \in X\}$ is guarded for Z w.r.t. to $\mathcal{H}$.

We use the term **linearly guarded** to indicate guarding w.r.t. to the class of all linear classifiers.

4 Iterative Nullspace Projection

Given a set of vectors $x_i \in \mathbb{R}^d$ and a set of corresponding discrete¹ protected attributes $z_i \in \mathcal{Z}$, we seek a *linear guarding function* g that remove the linear dependence between $\mathcal{Z}$ and $\mathcal{X}$.

We begin with a high-level description of our approach. Let c be a trained linear classifier, parameterized by a matrix $W \in \mathbb{R}^{k \times d}$, that predicts a property z with some accuracy. We can construct a projection matrix P such that $W(Px) = 0$ for all x , rendering W useless on dataset $\mathcal{X}$. We then iteratively train additional classifiers W' and perform the same procedure, until no more linear information regarding $\mathcal{Z}$ remains in X . Constructing P is achieved via nullspace projection, as described below. This method is the core of the INLP algorithm (Algorithm 1).

Nullspace Projection The linear interaction between W and a new test point x has a simple geometric interpretation: x is projected on the subspace spanned by W 's rows, and is classified according to the dot product between x and W 's rows, which is proportional to the components of x in the direction of W 's rowspace. Therefore, if we zeroed all components of x in the direction of W 's row-space, we removed all information used by W for prediction: the decision boundary found by the classifier is no longer useful. As the orthogonal component of the rowspace is the nullspace, zeroing those components of x is equivalent to projecting x on W 's nullspace. Figure 2 illustrates the idea for the 2 dimensional binary-classification setting, in which W is just a 2-dimensional vector.

For an algebraic interpretation, recall that the null-space of a matrix W is defined as the space $N(W) = \{x | Wx = 0\}$. Given the basis vectors of $N(W)$ we can construct a projection matrix $P_{N(W)}$ into $N(W)$, yielding $W(P_{N(W)}x) = 0 \ \forall x$.

This suggests a simple method for rendering z linearly guarded for a set of vectors X : training a linear classifier that is parameterized by W_0 to predict Z from X , calculating its nullspace, finding the orthogonal projection matrix $P_{N(W_0)}$ onto the nullspace, and using it to remove from X those components that were used by the classifier for predicting Z .

¹While this work focuses on the discrete case, the extension to a linear regression setting is straightforward: A projection to the nullspace of a linear regressor w enforces $wx = 0$ for every x , i.e., each input is regressed to the non-informative value of zero.